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Ml LAb-8

Aim:- To apply principle Component Analysis for dimensionality reduction and visualize the transformed feature space

```
# step 1 : Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
import seaborn as sns
```

```
iris = load_iris()
X = iris.data
y = iris.target
```

```
scalar = StandardScaler()
X_scaled = scalar.fit_transform(X)
```

```
print(X_scaled)
```

Show hidden output

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
```

```
print(X_pca)
```

Show hidden output

```
print("Explained Variance Ratio:",
      pca.explained_variance_ratio_)
```

```
Explained Variance Ratio: [0.72962445 0.22850762]
```

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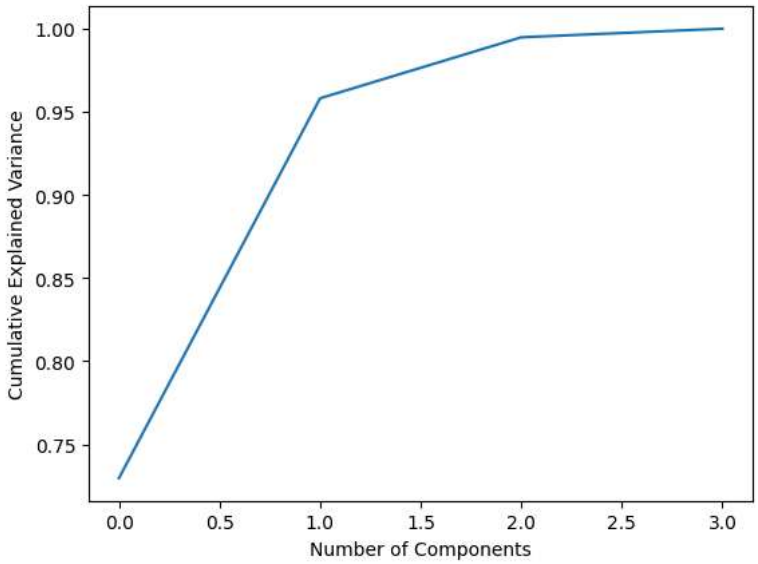
```
pca = PCA()
pca.fit(X_scaled)
plt.plot(np.cumsum(pca.explained_variance_ratio_))
plt.xlabel('Number of Components')
plt.ylabel('Cumulative Explained Variance')
plt.show()
```



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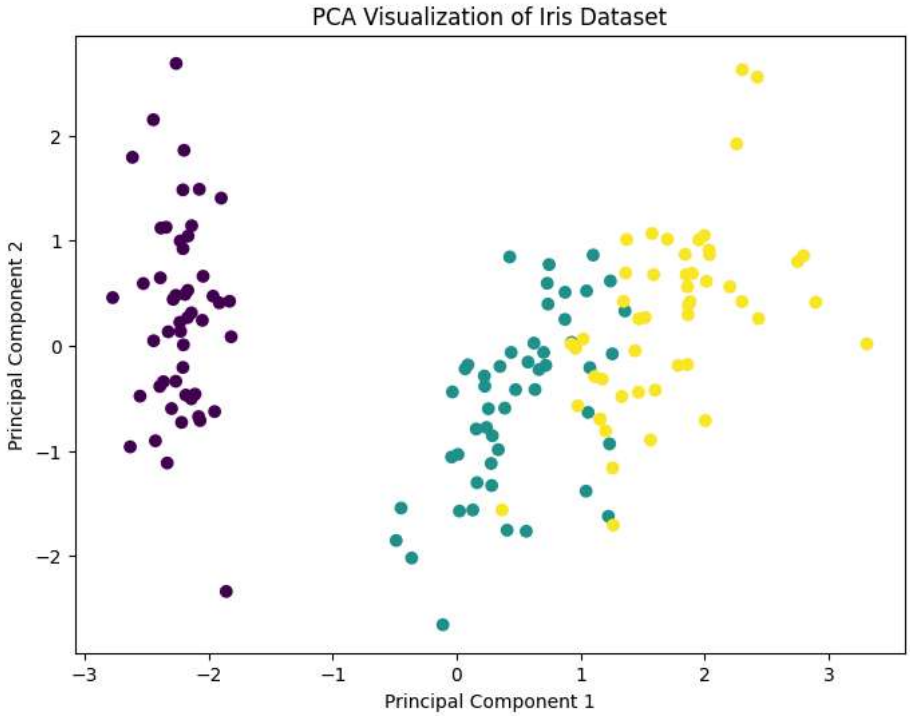
```
plt.figure(figsize=(8, 6))
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('PCA Visualization of Iris Dataset')
```

```
Text(0.5, 1.0, 'PCA Visualization of Iris Dataset')
```



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